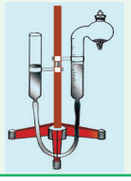
Activity

Take a test tube with some germinated seeds and fill with water. Keep this test tube after some time until liberation of CO_2. When the carbon dioxide from respiration is mixed to water, carbonic acid (H_2CO_3) is produced. Therefore, as more carbon dioxide is released, the solution becomes more acidic. You will see changes in pH as an indicator using blue litmus paper changed into red that respiration has occurred

$$CO_2 + H_2O \longrightarrow H_2CO_3$$

The apparatus used for determining respiration and RQ is called Ganong's Respirometer.



Respiratory quotients of some other substances

Proteins	:	0.8–0.9
Oleic acid (Fat)	:	0.71
Palmitic acid (Fat)	:	0.36
Tartaric acid	:	1.6
Oxalic acid	:	4.0

14.7 Anaerobic Respiration

14.7.1 Fermentation

Some organisms can respire in the absence of oxygen. This process is called **fermentation or anaerobic respiration** (Figure 14.12). There are three types of fermentation:

1. Alcoholic fermentation
2. Lactic acid fermentation
3. Mixed acid fermentation

1. Alcoholic fermentation

The cells of roots in water logged soil respire by alcoholic fermentation because of lack of oxygen by converting pyruvic acid into ethyl alcohol and CO_2. Many species of yeast (*Saccharomyces*) also respire anaerobically. This process takes place in two steps:

(i) $2CH_3COCOOH \xrightarrow[TPP]{\text{Pyruvate decarboxylase}} 2CH_3CHO + 2CO_2\uparrow$
Pyruvic acid → Acetaldehyde

(ii) $2CH_3CHO + 2NADH + 2H^+ \xrightarrow{\text{Alcohol dehydrogenase}} 2CH_3CH_2OH + 2NAD^+$
Acetaldehyde → Ethyl alcohol

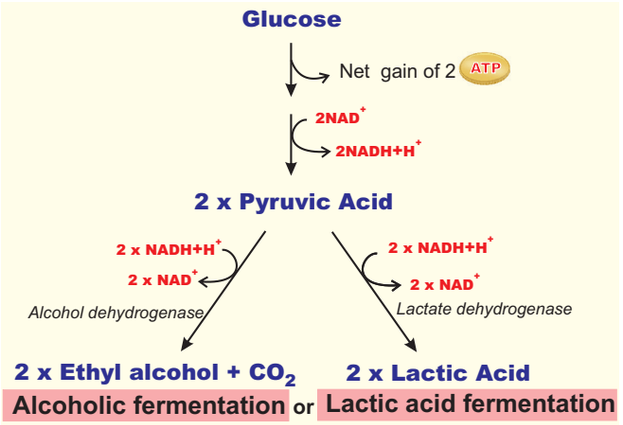


Figure 14.12: Anaerobic Respiration

Table 14.3: Comparison of alcoholic fermentation and lactic acid fermentation

Alcoholic fermentation	Lactic acid fermentation
1. It produces alcohol and releases CO_2 from pyruvic acid.	It produces lactic acid and does not release CO_2 from pyruvic acid.
2. It takes place in two steps.	It takes place in single step.
3. It involves two enzymes, pyruvate decarboxylase with Mg^{++} and alcohol dehydrogenase.	It uses one enzyme, lactate dehydrogenase with Zn^{++}.
4. It forms acetaldehyde as intermediate compound.	Does not form any intermediate compound.
5. It commonly occurs in yeast.	Occurs in bacteria, some fungi and vertebrate muscles.

XIth_Botany_Chapter_14_EM.indd 301 5/17/2020 9:21:01 PM